

INMO(15th) – 2000

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. The incircle of a triangle ABC touches BC, CA, AB in K, L and M respectively. The line through A and parallel to LK meets MK in P and the line through A and parallel to MK meets Lm in Q. Show that the line PQ bisects the sides AB and AC of the triangle ABC.
 2. Solve for integers x, y, z: $x + y = 1 - z$, $x^3 + y^3 = 1 - z^2$.
 3. If a, b, c, x are real numbers such that $abc \neq 0$ and $\frac{xb + (1-x)c}{a} = \frac{xc + (1-x)a}{b} = \frac{xa + (1-x)b}{c}$ then prove that $a = b = c$.
 4. In a convex quadrilateral PQRS, $PQ = RS$, $(\sqrt{3} + 1)QR = SP$ and $\angle RSP - \angle SPQ = 30^\circ$, prove that $\angle PQR - \angle QRS = 90^\circ$.
 5. Let a, b, c be three real numbers such that $1 \geq a \geq b \geq c \geq 0$. Prove that if λ is a root of the cubic equation $x^3 + ax^2 + bx + c = 0$ (real or complex), then $|\lambda| \leq 1$.
 6. For any natural numbers n, ($n \geq 3$), let f(n) denote the number of non-congruent integer sided triangles with perimeter n (Example: $f(3) = 1$, $f(4) = 0$, $f(7) = 2$). Show that:
 - (a) $f(1999) > f(1996)$
 - (b) $f(2000) > f(1997)$.

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